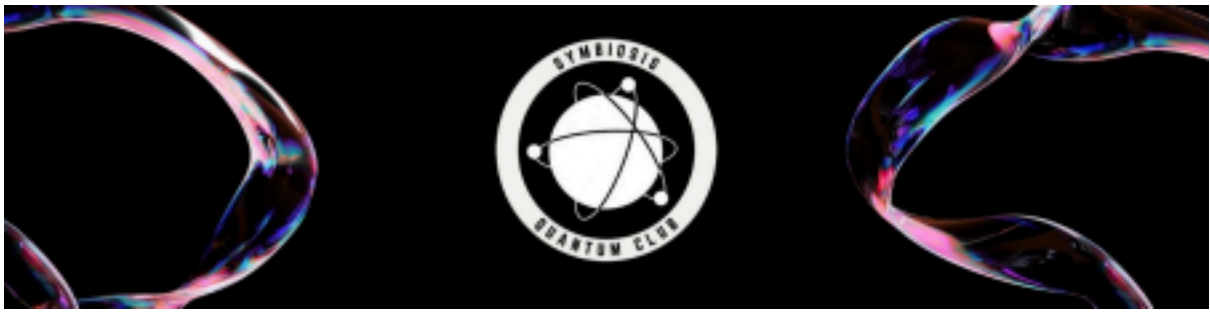




Symbiosis Quantum Club (SQC)

[LinkedIn](#) | [Instagram](#) | [GitHub](#) | quantumclub@sitpune.edu.in

The *Symbiosis Quantum Club (SQC)* is an initiative of the Department of Artificial Intelligence and Machine Learning (AIML) at Symbiosis Institute Of Technology, Pune.

This club aims to explore the fascinating intersection of quantum computing and artificial intelligence, providing a platform for students to engage in cutting-edge research, discussions, and projects.

1. Introduction:

Who are We?

A dedicated group of students at SIT focused on building and developing in the field of Quantum Computing. We work alongside our faculty, research organisations (such as CDAC Pune), and startups working in the Quantum Computing Space. *Our primary goal is to develop a robust learning community of students, building the future of Quantum Computing Research and Development in Pune.* Since our founding in June 2025, we have hosted interactive workshops, guest lectures, and hands-on demonstrations every month.



Our Team:

- **Faculty Mentor:** Dr. Archana Y. Chaudhari
- **President Advisory:** Samarth Bhadane
- **Vice President Advisory:** Disha Gupta
- **RnT Advisory (Research):** Anirudh Ganapathy Raman
- **RnT Advisory (Technical):** Eric Siqueira
- **President:** Riddhima Deshmukh
- **Vice President:** Akarsh Sunil
- **Treasurer:** Ishan Malviya
- **Research and Technical Head:** Shayan Bhowmik
- **Events Head:** Shantanu Shaji
- **Events Co-Head:** Apurva Satkar
- **Media and Documentation Head:** Sarvesh Dungarwal
- **Media and Documentation Co-Head:** Jeet Paghdar
- **Website Lead:** Ansh Saini

2. Motivation Behind Our Club:

The Department of AI&ML, SIT Pune, in association with CDAC (Centre for Development of Advanced Computing), Pune, hosted the [Quantum Accelerated Computing Symposium and Workshop](#), which was held on 15th and 16th October 2024. A two-day event offered an exclusive look into the future of quantum computing and its potential to transform fields like AI and machine learning.

This event sparked a wave of interest and enthusiasm by AIML Students for building in the space of Quantum Computing and Quantum Machine Learning. Supported by our Faculty-in-charge, [Dr. Archana Chaudhari](#) and Head of AIML Department [Dr. Shruti Patil](#), We founded the Symbiosis Quantum Club in June 2025, to further growth.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

**Quantum Accelerated Computing
Symposium and Workshop**



**15th-16th
October'2024**

Chief Guests


Ms. Lakshmi Parat
(Convener)
Program Director,
AI-QT, C-DAC Pune


Aditya Sinha
Director, C-DAC
Patna


Dr. U B Desai
Prof. Emeritus & Founding
Director, IIT Hyderabad
Chancellor, Anug
University


Col (Retd.) P. Sol Shankar
VSM, Director
(Signal), Indian
Army


Dr. Ketan Katscha
Dean & Director
IIT Pune

Speakers


Dr. Shruti Patil
HOD,
Dept. Of AIML

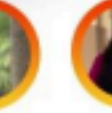

Dr. Kalyan Dasgupta
Research Staff
Member,
IBM, India


Prof. Anjan Chakrabarti
Professor and Director, A.K.
Choudhury School of IT,
University of Calcutta &
Adj. Prof. IIT Delhi


Prof. Arindam Ghosh
Professor
IISc Bangalore


Dr. Santu Sarder
Director,
QVSI-QT Lab


Mr. Arpit Saxena
Scientist-S,
C-DAC Pune


Prof. Joyee Ghosh
Associate Professor
IIT Delhi


Dr. Vidya Niranjan
Professor & HOD,
Biotechnology,
RV College of Engineering,
Bangalore



3. Events We Organized:

Qiskit Fallfest 2025, 8th & 11th October, 2025:

In collaboration with the Department of Artificial Intelligence and Machine Learning, the club organized Qiskit Fallfest. The event featured an online expert session led by Mrs Kavitha Yogaraj, Qiskit Advocate, followed by an offline student-led workshop. The students explored concepts like superposition and quantum gates, to then build and simulate quantum circuits using Qiskit. Overall the workshop of Qiskit Fall Fest

2025 successfully bridged the gap between theory and practice in quantum computing. Through live coding, visual demonstrations, and peer-led guidance, participants experienced how abstract quantum concepts come to life using Qiskit.





The Quantum Icebreaker, 25th July 2025:

Our first session was a beginner-friendly introduction to quantum computing concepts. This was an interactive session, to explore the rapidly developing world of Quantum Computing with our esteemed alumni. Our Guest Speaker, Mr. Om Shankar was a Quantum AI Intern in Tech Mahindra, explained his experiences researching in the field of Quantum AI and its implementations in Quantum Biology.



Agentic AI X Quantum Agents, 23rd August, 2025:

This was a student-led, collaborative workshop, which equipped Junior and Sophomore Students with hands-on, industry-relevant exposure pertaining to the emerging intersection of quantum computing and AI. Particularly through **Quantum Agents**; autonomous, intelligent systems empowered by quantum-enhanced decision-making, planning, and optimization.

We explored document intelligence (real-time processing using advanced LLMs), built a RAG-based AI chatbot with multilingual integration, and developed an industry-standard agentic AI project. Additionally, we discussed how quantum-accelerated agent architectures are shaping the next-generation of AIML research, enabling faster and more adaptive systems.



Qiscade [Quantum Arcade], 19th September, 2025:

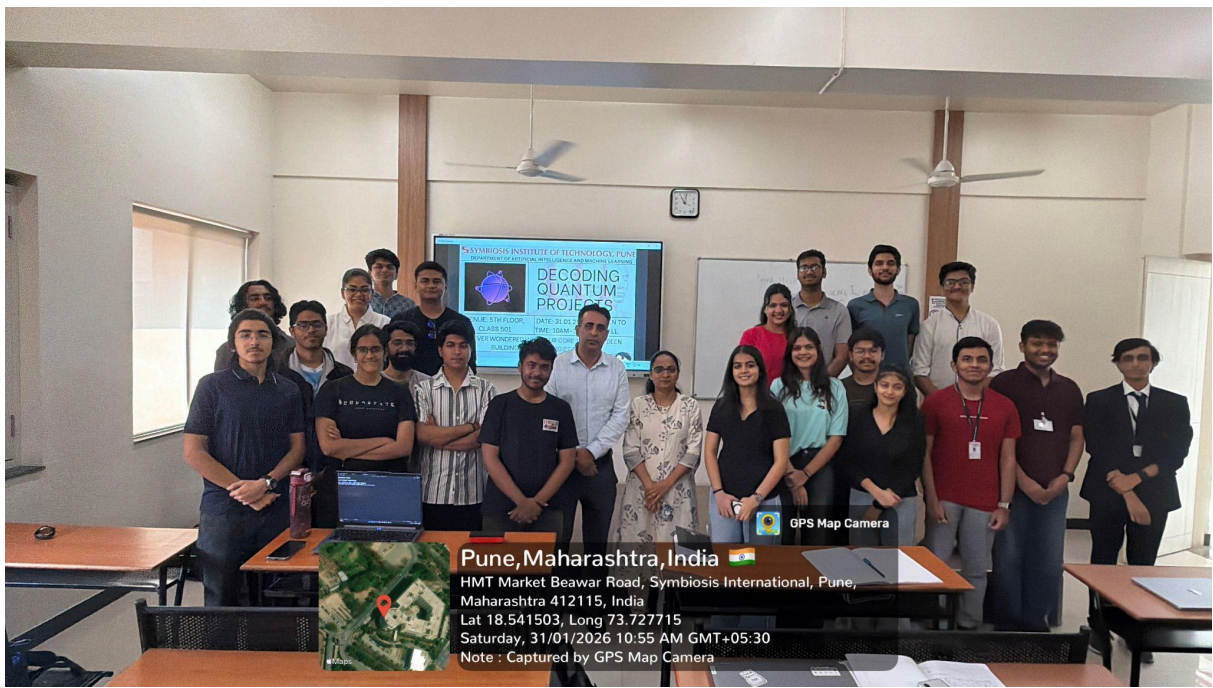
As part of Symbitech, the college's premier techfest, the club hosted a quantum-inspired arcade game developed in-house. The game introduced participants to fundamental quantum concepts through interactive gameplay. Players navigated through levels with puzzles based on ideas such as superposition and entanglement. The event attracted over 200 participants and saw an overwhelmingly positive response. The event was considered a success in engaging and spreading awareness about Quantum concepts on the campus.





Decoding Quantum Projects, 31st January, 2026:

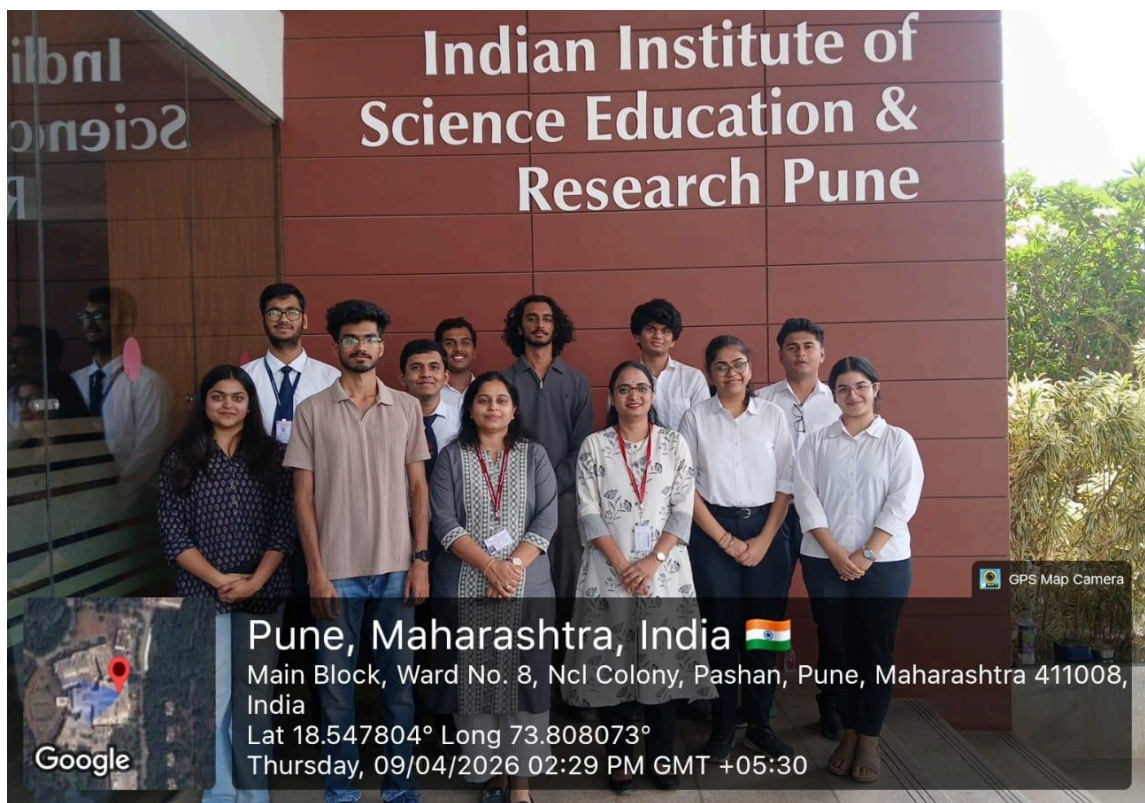
This interactive project discussion session showcased ongoing projects in quantum computing and the research challenges they addressed. It was led by the pre-final year members of the club. Participants gained exposure to several applications of quantum technologies and explored emerging research avenues. The club engaged in an insightful discussion through an extensive Q&A session post the presentations. The event gave the students a clear understanding of the scope and complexity of research and explored several opportunities available in quantum computing.



Industrial visit to I-Hub Quantum Technology Foundation, IISER Pune , 9th April, 2026:

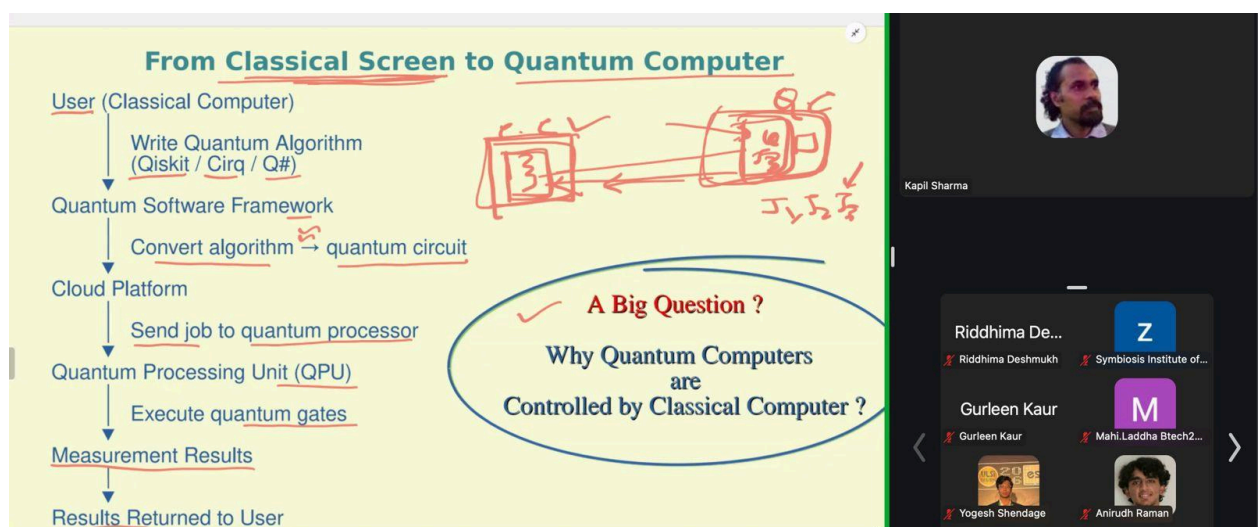
The club organized an industrial visit to i-Hub QTF, where participants gained first-hand exposure to cutting-edge quantum technologies through expert-led workshops and laboratory tours. We explored hands-on Quantum Cryptography and Quantum Optics kits learning the fundamentals of quantum key distribution and optical systems. The staff also explored various internal projects the researchers were working on at the time, including the development of a 20 qubit ion trap quantum processor and transportable atomic gravimeter, and other next-gen technology.

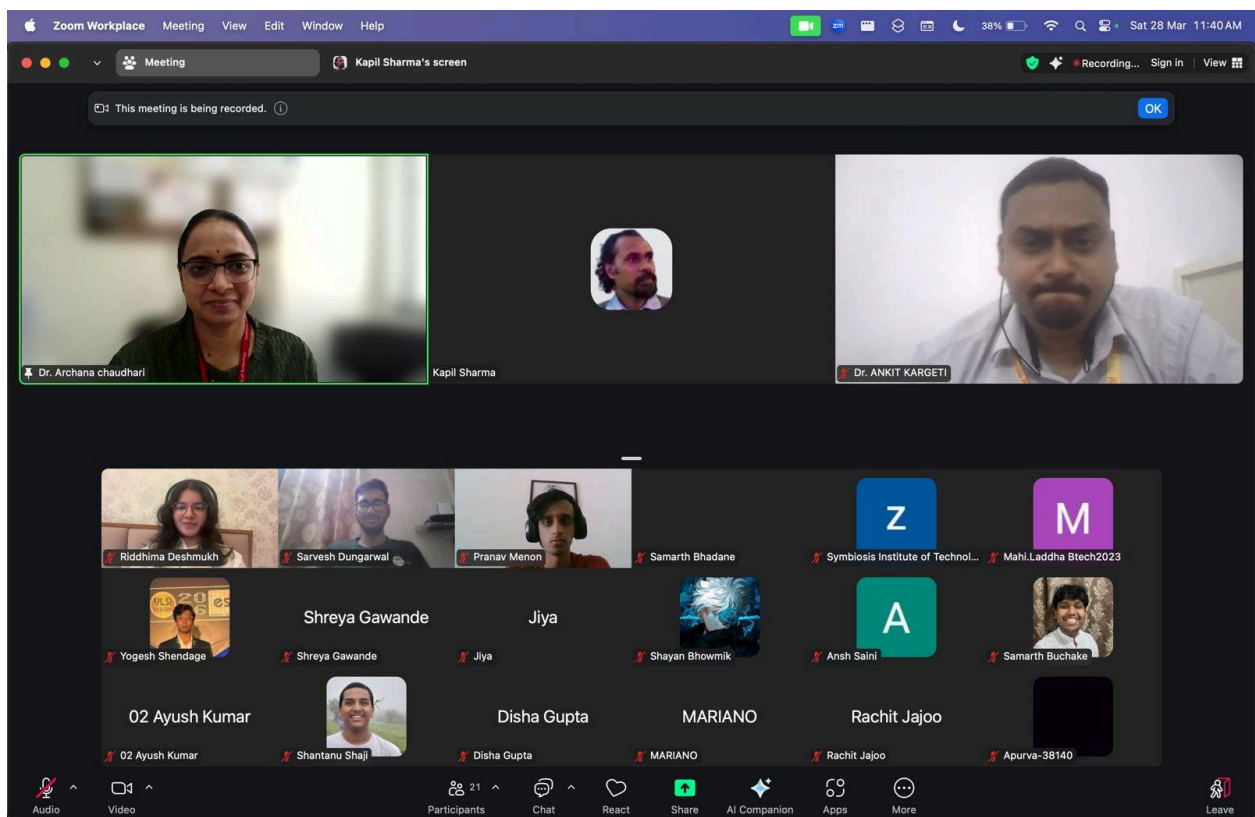




Emerging Trends in Quantum Computing, 28th March, 2026:

Delivered by Dr. Kapil Sharma, this expert session gave students an in-depth overview of the evolving quantum computing landscape. Many topics including quantum hardware architecture, hybrid quantum-classical systems, and quantum AI, and quantum cryptography were discussed. The session highlighted India's National Quantum Mission and career opportunities in quantum. The session was followed by an engaging discussion about the future of quantum technology where students displayed their curiosity and enthusiasm for their journey with quantum technology.





Workshop on National Science Day, 27th February, 2026:

On the occasion of National Science Day, the club conducted a hands-on Power BI workshop for students of Symbiosis International School at SIT Pune. Using real-world datasets, participants learned the basics of data processing and visualization, including data cleaning, transformation and dashboard creation. The participants were taught how data can tell stories, and how visualizations are the bridge between data and actually understanding it. The event was very engaging and the theme of movies kept the entire session very entertaining.





4. Technical Projects Overview:

A brief overview of our completed research projects:

- Project 1: **Quant-D: Interactive Visualization of Single-Qubit Transformations for Quantum Computing Education and Research**
Done by our club member, **Anirudh Raman**.

FRL-I AY 2024-25		SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE	
TITLE: Quant-D: A GUI-based Quantum Circuit Simulator using Qiskit Names of the Students: Ganapathy Raman Anirudh & Aran Tati Name of the Guide: Dr. Reshika Agrawal			
INTRODUCTION Quantum computing education is often restricted to theoretical study or expensive simulators that require coding experience. Beginners and students struggle to visualize how quantum gates affect qubit states. Moreover, the impact of real-world noise on qubit behavior is rarely demonstrated in an interactive manner. Therefore, there is a need for an easy-to-use platform that allows intuitive, visual exploration of quantum circuits and noise models without requiring deep prior knowledge of programming or quantum mechanics.	OUTCOME of the Project Work Built a functional GUI-based quantum circuit simulator for single-qubit operations. Enabled real-time visualization of qubit state evolution and Bloch sphere transformations. Demonstrated the effects of real-world quantum noise (depolarizing, amplitude damping, phase damping). Made quantum circuit construction and simulation accessible to users without coding knowledge. Provided QASM export functionality, making circuits compatible with external quantum simulators. Enhanced understanding of quantum gate operations, state vectors, and error impacts among users.	Tools Used Python 3.x: Programming language. Flask: GUI framework for building the application interface. Qiskit: Library for quantum computing simulations and circuit management. Matplotlib: For 3D Bloch sphere visualizations. Qiskit Aer Simulator: For simulating quantum noise and measurements.	
OBJECTIVES & AIMS 1) Create a GUI-based single-qubit circuit simulator. 2) Allow visualization of state evolution and Bloch sphere. 3) Simulate quantum noise (Depolarizing, Amplitude Damping, Phase Damping). 4) Export QASM code for further use.	METHODOLOGY 1) Create a GUI-based single-qubit circuit simulator. 2) Allow visualization of state evolution and Bloch sphere. 3) Simulate quantum noise (Depolarizing, Amplitude Damping, Phase Damping). 4) Export QASM code for further use.	References [1] IBM Quantum, "Qiskit: An open-source framework for quantum computing," [Online]. Available: https://qiskit.org/documentation [2] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary Edition, Cambridge University Press, 2010. [3] IBM Quantum Lab, "Experience quantum computing on the IBM Cloud," [Online]. Available: https://quantum-computing.ibm.com [4] Matplotlib Developers, "Matplotlib: Visualization with Python," [Online]. Available: https://matplotlib.org/ [5] Qiskit Aer, "High-performance quantum simulators for Qiskit," [Online]. Available: https://qiskit.org/documentation/qiskit/aer.html	
ACKNOWLEDGEMENT We would like to express our sincere gratitude to Prof. Reshika Agrawal, Department of Computer Science and Engineering, for her invaluable guidance, encouragement, and support throughout the course of this project. Her insights and consistent motivation played a pivotal role in shaping our understanding and refining the outcomes of our work. It was under her mentorship that we were able to approach the project systematically and achieve our objectives effectively. We are truly thankful for her dedication and mentorship.			

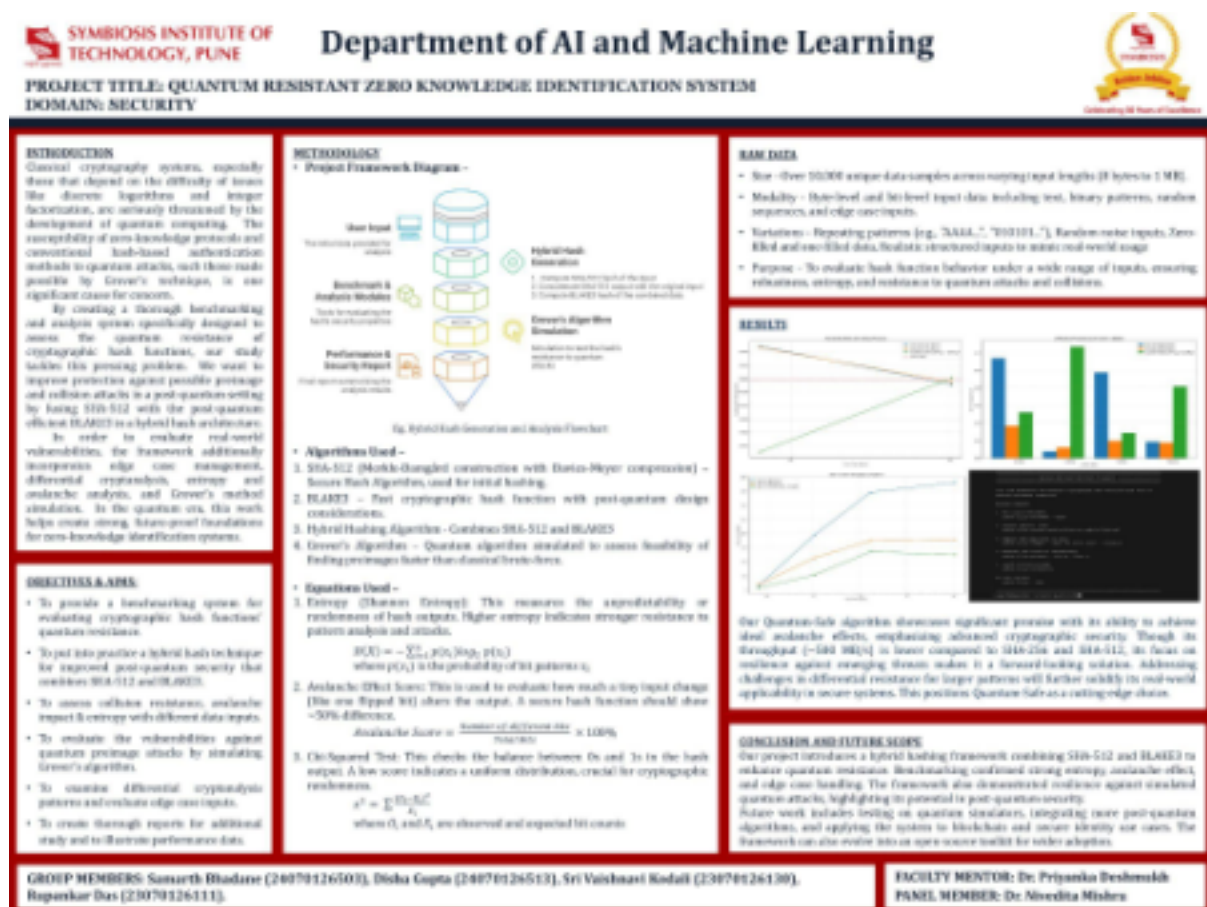
GitHub Link:

https://github.com/AnirudhRaman3277/Qubit_Visualization_Qiskit

About: This project expands a quantum visualization tool for plotting qubit states on the Bloch sphere, adding features like QASM export and noise models to observe decoherence in quantum circuits. Built with **Qiskit**, **Tkinter**, and **Matplotlib**, it aims to provide students and researchers with a user-friendly GUI that supports offline QASM export and unique noise-based visualization for better understanding of quantum computing concepts

● Project 2: Quantum Resistant Zero-Knowledge Identification System

Done by our club members, **Samarth Bhadane**, **Disha Gupta**.



GitHub Link: <https://github.com/samarthsb4real/Quantum-Resistant-ZKI.git>

About: The project uses a post-quantum secure identity protocol design to resist quantum adversaries. The system integrates **SHA-512** and **BLAKE3** hashing with **Zero-Knowledge Proofs** to ensure privacy-preserving and verifiable authentication. Its emphasis was placed on lightweight, configurable design, making it adaptable to diverse systems without requiring specialized quantum hardware. The framework demonstrates a practical step toward quantum-resistant cryptographic authentication.

- Project 3: **QuantumStego**

Done by our club member, **Disha Gupta**

About: Quantum Stego is a secure steganography platform that allows users to hide secret messages inside images, secured using simulated Quantum Key Distribution. The encryption ensures that even if the stego image is intercepted, the hidden message remains safe without the correct quantum-generated key. Deployed Link: <https://quantum-stego-iqyc.vercel.app/>

Additionally, we built a comprehensive **Quantum Computing Fundamentals Roadmap**, to teach new club members the fundamentals of quantum computing and its programming frameworks (*Qiskit, Cirq, Tensorflow-Quantum, PennyLane*). Starting from fundamental Quantum Computing Algorithms, we leveraged and showcased real life use-cases of quantum computing in the fields of Cybersecurity, Finance, etc.

GitHub Link:

<https://github.com/Symbiosis-Quantum-Club/Quantum-Computing-and-QML-a-Comprehensive-Roadmap>

5. Our Goals:

1. Community Building: **From 8 Founding Members, we have built a Quantum Computing Learning Community of 138 Students.** These students participate in our monthly workshops, discuss developments in the field and ideate on future research opportunities.
2. Skill Development: Building a college coding culture of Certification training, developing research publications, teaching industry level skills in Qiskit, Cirq, PennyLane, DWave-Ocean, and other such Quantum frameworks.
3. Research & Outcomes: Publish our research projects in reputed academic journals, participate in hackathons, contribute to open-source QC tools. 4. Industry Collaboration: Secure internships in Quantum Computing Companies, establish MoUs for research partnerships, inviting subject matter experts for guest lectures.

6. In Conclusion,

At the Symbiosis Quantum Club, our journey sparked by academic curiosity and driven by industry relevance, has grown into a vibrant, hands-on exploration of Quantum Computing. From interactive workshops and industry-speaker sessions to ambitious research projects and upcoming gamified events, we're building both knowledge and community. **In doing so, we envision SQC not merely as a student initiative, but as a catalyst for Pune's quantum computing ecosystem.**

Thank You.